



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

FINAL EXAMINATION – FALL 2022

COURSE NAME:

Big Data Analytics

COURSE NUMBER:

SOC4170

EXAMINATION DATE:

TIME:

EXAMINATION DURATION:

90 Minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

Calculator

SPECIAL INSTRUCTIONS:

- Answers will only be evaluated if they are readable.
- Show all your work. For some questions, you may get partial credit even if the end result is wrong due to a calculation mistake. If you make assumptions, state your assumptions clearly and precisely.

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

CALCULATORS ARE ALLOWED.

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

1.[10 points] (True/False questions) Write T if the following statement is True, write F otherwise inside the table below

a.	b.	c.	d.	e.	f.	g.	h.	i.	j.
F	T	T	T	T	T	T	F	F	F

- a. In credit card fraud analysis, precision is the most valuable metric.
- b. When using dropout, half of hidden layer neurons are disabled before the start of every mini-batch.
- c. delta means how much small change of z value causes small change of cost function.
- d. In Hadoop, data transfer is handled implicitly by HDFS.
- e. We can reduce overfitting by reducing the size of the network.
- f. In cross validation, we randomly partition the data into k mutually exclusive subsets.
- g. Frequent pattern mining can be used in cluster analyses.
- h. In frequent patterns, support is usually bigger than confidence.
- i. When weight values are very small, L2 regularization decreases weights more than L1 regularization
- j. The way of sampling input tuples for bagging is the same as the way for mini-batch.

2.[14 points] Least Squares (Linear Regression) Here is a dataset given as (x, y) and the cost function C:

$$\{(1, 3), (3, 5), (4, 6), (5, 8), (7, 10)\} \quad C = \sum_{i=1}^n (y_i - (ax_i + b))^2$$

(a) [8 Points] Write 2 linear equations in terms of 2 variables a and b to minimize the cost. Show all your work

$$\frac{\partial C}{\partial a} = 2 \sum (y_i - (ax_i + b)) (-x_i) = 0$$
$$\sum x_i^2 \times a + \sum x_i \times b = \sum x_i y_i$$

$$\frac{\partial C}{\partial b} = 2 \sum (y_i - (ax_i + b)) (-1) = 0$$
$$\sum x_i \times a + \sum b = \sum y_i$$

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

(b) [4 Points] What are the values of the variables? Show all your work

$$\begin{aligned} 100a + 20b &= 152 \\ 20a + 5b &= 32 \\ 100a + 25b &= 160 \end{aligned} \quad \begin{aligned} 5b &= 8 & b &= \frac{8}{5} = 1.6 \\ 20a &= 24 & a &= 1.2 \end{aligned}$$

(c) [2 Points] What is the predicted value of y when x = 10?

$$y = 13.6$$

3.[8 points] BP4

(a) [8 Points] Prove BP3 in question 5.

$$\begin{aligned} \frac{\partial C}{\partial b_j^l} &= \sum_k \frac{\partial C}{\partial z_k^l} \frac{\partial z_k^l}{\partial b_j^l} \\ z_k^l &= \sum_i w_{ki}^l a_i^{l-1} + b_k^l \\ \text{if } k=j &\text{ then } \frac{\partial z_k^l}{\partial b_j^l} = 1 \\ \frac{\partial C}{\partial b_j^l} &= \frac{\partial C}{\partial z_j^l} = \delta_j^l \end{aligned}$$

[18 points] Adaboost

Given a set of d class-labeled tuples, $(\mathbf{X}_1, y_1), \dots, (\mathbf{X}_d, y_d)$

Initially, all the weights of tuples are set the same $(1/d)$

Let $d = 4$

and here is the sequence of your random numbers for sampling $D_1: (x_1, x_2, x_3, x_3)$

From D_1 , your model M_1 is generated. M_1 predicted the class labels of x_2 and x_3 correctly but misclassified x_1

(a) [2 Points] What is the error (M_1)? Refer to the following equation and show all your work.

$$\text{error}(M_i) = \sum_j^d w_j \times \text{err}(\mathbf{X}_j)$$

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

$$\text{error}(M_1) = \frac{3}{4}$$

(b) [4 Points] What are the weights of 4 tuples $(x_1 - x_4)$ before normalization? Show all your work

$$w_1 = \frac{1}{4} \quad w_2 = w_3 = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12} \quad w_4 = \frac{1}{4}$$

(c) [4 Points] Write the sum of old weights and the sum of new weights. And write the weights of 4 tuples $(x_1 - x_4)$ after normalization? Show all your work

$$\begin{aligned} \text{sum of old} &= 1 & \text{sum of new} &= \frac{1}{4} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{1}{2} \\ w_1 &= \frac{1}{2} & w_2 = w_3 &= \frac{1}{6} & w_4 &= \frac{1}{4} \end{aligned}$$

In the next round, the sequence of your random numbers for sampling D_2 : $(x_1 x_1 x_2 x_4)$

From D_2 , your model M_2 is generated. M_2 predicted the class labels of x_1 and x_2 correctly but misclassified x_4

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

(d) [4 Points] What are the weights of 4 tuples ($x_1 - x_4$) before normalization? Show all your work

$$w_1 = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6} \quad w_2 = \frac{1}{6} \times \frac{1}{3} = \frac{1}{18}$$

$$w_3 = \frac{1}{6} \quad w_4 = \frac{1}{4}$$

(e) [4 Points] Write the sum of old weights and the sum of new weights. And write the weights of 4 tuples ($x_1 - x_4$) after normalization? Show all your work

m

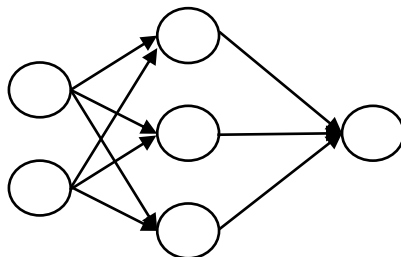
$$\text{Sum of old: } \frac{1}{2} + \frac{1}{2} + \frac{1}{6} + \frac{1}{4} = \frac{6+6+2+3}{12} = \frac{17}{12}$$

$$\text{Sum of new: } \frac{1}{6} + \frac{1}{6} + \frac{1}{18} + \frac{1}{4} = \frac{6+6+2+9}{36} = \frac{23}{36}$$

$$w_1 = \frac{1}{6} \times \frac{51}{23} = \frac{17}{46} \quad w_2 = \frac{1}{18} \times \frac{51}{23} = \frac{17}{138} \quad w_3 = \frac{1}{6} \quad w_4 = \frac{51}{92}$$

5.[50 points] Feedforward and back propagation

Below is a neural network:



All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} -1 & 0 \\ -1 & 1 \\ -1 & 1 \end{bmatrix} \quad w^3 = \begin{bmatrix} -1 & 1 & -1 \end{bmatrix} \quad b^2 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \quad b^3 = [1]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

$$y_1 = [1] \quad y_2 = [1] \quad y_3 = [1] \quad y_4 = [1]$$

And the learning rate is $\eta = 0.4$

We are processing 4 inputs at the same time by having all 4 of them as a matrix.

Round off all the number to the nearest 1000th. For example, 6.6543 is 6.654 and 3.3355 is 3.336.

And the back propagation equations are as follows:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (\text{BP1})$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (\text{BP2})$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (\text{BP3})$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad (\text{BP4})$$

(a) [4 Points] What is Z^2 matrix? First represent it with matrix format and write the result. Show all your work.

$$Z^2 = \begin{bmatrix} -1 & 0 \\ -1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} -1 & -1 & -1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 & -2 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 0 & -2 & -1 \end{bmatrix}$$

(b) [4 Points] What is a^2 matrix? Show all your work.

$$a^2 = \begin{bmatrix} 0.269 & 0.269 & 0.119 & 0.119 \\ 0.731 & 0.881 & 0.5 & 0.731 \\ 0.269 & 0.5 & 0.119 & 0.269 \end{bmatrix}$$

z	0	0.5	1	1.5	2	-0.5	-1	-1.5	-2
a	0.5	0.622	0.731	0.818	0.881	0.378	0.269	0.182	0.119

z	1.112	1.193	1.220	1.248	1.262	1.331	1.338	1.343	1.518
a	0.753	0.767	0.772	0.777	0.779	0.791	0.792	0.793	0.820

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

$$Z^3 = [-1 \ 1 \ -1] a^2 + [1 \ 1 \ 1 \ 1] \\ = [1.193 \ 1.112 \ 1.262 \ 1.343]$$

(d) [4 Points] What is a^3 matrix? Show all your work.

$$a^3 = [0.767 \ 0.753 \ 0.779 \ 0.793]$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

$${}^1\delta^3 = (0.767 - 1) \times 0.767 \times (1 - 0.767) = -0.042$$

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

$${}^2\delta^3 = (0.753 - 1) \times 0.753 \times (1 - 0.753) = -0.046$$

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

$${}^3\delta_1^3 = (0.779 - 1) 0.779 (1 - 0.779) = -0.038$$

(h) [3 Points] What is ${}^4\delta^3$ value? ${}^4\delta^3$ is the layer 3 delta for input 4. Show all your work.

$${}^4\delta_1^3 = (0.793 - 1) 0.793 (1 - 0.793) = -0.034$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

$$\frac{\partial C}{\partial b^3} = (-0.042 - 0.046 - 0.038 - 0.034) / 4 = -0.04$$

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

(j) [12 Points] What is average $\frac{\partial C}{\partial w^3}$ matrix? Show all your work.

$$\begin{aligned}\frac{\partial C}{\partial w_{11}^3} &= (-0.269 \times 0.042 - 0.269 \times 0.046 - 0.119 \times 0.038 \\ &\quad - 0.119 \times 0.034) / 4 = -0.008 \\ \frac{\partial C}{\partial w_{12}^3} &= (-0.731 \times 0.042 - 0.881 \times 0.046 - 0.5 \times 0.038 \\ &\quad - 0.731 \times 0.034) / 4 = -0.029 \\ \frac{\partial C}{\partial w_{13}^3} &= (-0.269 \times 0.042 - 0.5 \times 0.046 - 0.119 \times 0.038 \\ &\quad - 0.269 \times 0.034) / 4 = -0.012\end{aligned}$$

(k) [3 Points] What is new b^3 vector? Show all your work.

$$b_1^3 = [1] + [0.4 \times 0.04] = [1.016]$$

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

(I) [3 Points] What is new w^3 matrix? Show all your work.

$$\begin{aligned} w^3 &= [-1 \quad 1 \quad -1] + \begin{bmatrix} 0.4 \times 0.008 & 0.4 \times 0.029 \\ & 0.4 \times 0.012 \end{bmatrix} \\ &= [-0.997 \quad 1.012 \quad -0.995] \end{aligned}$$